

# RaceDay — API Endpoint Plan

This table lists every endpoint the RaceDay REST API will expose in Part 2, planned before any application code is written. All routes are prefixed `/api`. Roles: **None** = public, **Any** = any logged-in user, **Organiser** / **Participant** = specific role required. Where an endpoint says "(owner)", the API must also check that the logged-in Organiser owns the Event/Category being modified.

## Authentication

| HTTP Method | Route              | Description                                                   | Role Required | Request Body                                     | Expected Response                                                                                                            |
|-------------|--------------------|---------------------------------------------------------------|---------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| POST        | /api/auth/register | Registers a new user as either an Organiser or a Participant. | None          | { fullName, email, password, role, phoneNumber } | 201 Created – new user (no password) returned. 400 Bad Request – validation failed. 409 Conflict – email already registered. |
| POST        | /api/auth/login    | Authenticates a user and issues a JWT.                        | None          | { email, password }                              | 200 OK – { token, role, userId }. 401 Unauthorized – invalid credentials.                                                    |

## User Profile

| HTTP Method | Route           | Description                                                                                      | Role Required | Request Body              | Expected Response                          |
|-------------|-----------------|--------------------------------------------------------------------------------------------------|---------------|---------------------------|--------------------------------------------|
| GET         | /api/users/me   | Returns the logged-in user's own profile.                                                        | Any           | None                      | 200 OK – user profile. 401 Unauthorized.   |
| PUT         | /api/users/me   | Updates the logged-in user's own profile details.                                                | Any           | { fullName, phoneNumber } | 200 OK – updated profile. 400 Bad Request. |
| GET         | /api/users/{id} | Returns a specific user's public profile (e.g. an Organiser viewing a Participant who enrolled). | Organiser     | None                      | 200 OK – user profile. 404 Not Found.      |

## Events

| HTTP Method | Route            | Description                                                                    | Role Required | Request Body                                          | Expected Response                                                     |
|-------------|------------------|--------------------------------------------------------------------------------|---------------|-------------------------------------------------------|-----------------------------------------------------------------------|
| GET         | /api/events      | Lists all upcoming events, so Participants can browse without logging in.      | None          | None                                                  | 200 OK – array of events.                                             |
| GET         | /api/events/{id} | Returns full detail for a single event, including venue and route information. | None          | None                                                  | 200 OK – event detail. 404 Not Found.                                 |
| POST        | /api/events      | Creates a new event.                                                           | Organiser     | { title, description, eventDate, eventType, venueId } | 201 Created – new event. 400 Bad Request.                             |
| PUT         | /api/events/{id} | Updates an event (owner).                                                      | Organiser     | { title, description, eventDate, eventType, venueId } | 200 OK – updated event. 403 Forbidden – not the owner. 404 Not Found. |
| DELETE      | /api/events/{id} | Deletes an event (owner).                                                      | Organiser     | None                                                  | 204 No Content. 403 Forbidden. 404 Not Found.                         |

## Categories

| HTTP Method | Route                            | Description                                                             | Role Required | Request Body                                    | Expected Response                                           |
|-------------|----------------------------------|-------------------------------------------------------------------------|---------------|-------------------------------------------------|-------------------------------------------------------------|
| GET         | /api/events/{eventId}/categories | Lists all categories (e.g. 5km, 10km, Half Marathon) for a given event. | None          | None                                            | 200 OK – array of categories. 404 Not Found.                |
| POST        | /api/events/{eventId}/categories | Adds a new category to an event (owner).                                | Organiser     | { name, distanceKm, entryFee, maxParticipants } | 201 Created – new category. 400 Bad Request. 403 Forbidden. |

|        |                      |                                             |           |                                                 |                                                          |
|--------|----------------------|---------------------------------------------|-----------|-------------------------------------------------|----------------------------------------------------------|
| PUT    | /api/categories/{id} | Updates a category (owner of parent event). | Organiser | { name, distanceKm, entryFee, maxParticipants } | 200 OK – updated category. 403 Forbidden. 404 Not Found. |
| DELETE | /api/categories/{id} | Deletes a category (owner of parent event). | Organiser | None                                            | 204 No Content. 403 Forbidden. 404 Not Found.            |

Event Enrolments

| HTTP Method | Route                                   | Description                                                                        | Role Required | Request Body | Expected Response                                                                                                                   |
|-------------|-----------------------------------------|------------------------------------------------------------------------------------|---------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------|
| POST        | /api/categories/{categoryId}/enrolments | Enrols the logged-in Participant into a category, allocating a bib number.         | Participant   | None         | 201 Created – { enrolmentId, bibNumber }. 404 Not Found – category doesn't exist. 409 Conflict – already enrolled or category full. |
| GET         | /api/enrolments/my                      | Returns the logged-in Participant's own enrolments, past and upcoming.             | Participant   | None         | 200 OK – array of enrolments.                                                                                                       |
| GET         | /api/events/{eventId}/enrolments        | Returns all enrolments for an event, for the Organiser to manage entrants (owner). | Organiser     | None         | 200 OK – array of enrolments. 403 Forbidden. 404 Not Found.                                                                         |
| DELETE      | /api/enrolments/{id}                    | Cancels the logged-in Participant's own enrolment (owner).                         | Participant   | None         | 204 No Content. 403 Forbidden. 404 Not Found.                                                                                       |

Results

| HTTP Method | Route                                 | Description                                                                         | Role Required | Request Body                     | Expected Response                                                        |
|-------------|---------------------------------------|-------------------------------------------------------------------------------------|---------------|----------------------------------|--------------------------------------------------------------------------|
| POST        | /api/enrolments/{enrolmentId}/results | Captures a race-day result against an enrolment (owner of parent event).            | Organiser     | { finishTime, position, status } | 201 Created – new result. 400 Bad Request. 403 Forbidden. 404 Not Found. |
| PUT         | /api/results/{id}                     | Corrects/updates a captured result (owner of parent event).                         | Organiser     | { finishTime, position, status } | 200 OK – updated result. 403 Forbidden. 404 Not Found.                   |
| GET         | /api/results/my                       | Returns the logged-in Participant's personal performance history across all events. | Participant   | None                             | 200 OK – array of results.                                               |
| GET         | /api/events/{eventId}/results         | Returns the public results list for an event (e.g. a leaderboard).                  | None          | None                             | 200 OK – array of results. 404 Not Found.                                |